

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle \cdot, \cdot \rangle$ platí (pro libovolné $u, v \in V$)

$$a) \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2),$$

$$b) \operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2),$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

uvolňovací

pravidlo ... $\|\vec{u} - \vec{v}\|^2 = 2\|\vec{u}\|^2 + 2\|\vec{v}\|^2 - \|\vec{u} + \vec{v}\|^2$

$$a) \text{PS} \stackrel{\downarrow}{=} \frac{1}{4}(\|\vec{u} + \vec{v}\|^2 - 2\|\vec{u}\|^2 - 2\|\vec{v}\|^2 + \|\vec{u} + \vec{v}\|^2) =$$

$$= \frac{1}{2}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2)$$

$$\langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle;$$

$$\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$$

definice
normy

$$\|\vec{u} + \vec{v}\|^2 \stackrel{\downarrow}{=} \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle \stackrel{\downarrow}{=} \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle =$$

def. normy.

skalarizace skalar. s.

$$\stackrel{\downarrow}{=} \|\vec{u}\|^2 + \|\vec{v}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} =$$

$$z + \bar{z} = 2 \operatorname{Re}(z), \quad z \in \mathbb{C}$$

$$\stackrel{\downarrow}{=} \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

$$\text{PS} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 + 2 \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) - \|\vec{u}\|^2 - \|\vec{v}\|^2) = \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \text{LS}$$

$$\text{PS} = \text{LS}$$

b)

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u - iv\|^2 - \|u + iv\|^2)$$

def. normy;

$$\langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle; \quad \langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$$

$$\text{PS} \stackrel{\downarrow}{=} \frac{1}{4}(\langle \vec{u}, \vec{u} - i\vec{v} \rangle + \langle -i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u}, \vec{u} + i\vec{v} \rangle - \langle i\vec{v}, \vec{u} + i\vec{v} \rangle) =$$

def. normy;

$$\langle k\vec{u}, \vec{v} \rangle = k \langle \vec{u}, \vec{v} \rangle;$$

$$\langle \vec{u}, k\vec{v} \rangle = k \langle \vec{u}, \vec{v} \rangle$$

$$\stackrel{\downarrow}{=} \frac{1}{4}(\|\vec{u}\|^2 - i \langle \vec{u}, \vec{v} \rangle + i \langle \vec{v}, \vec{u} \rangle + i(-i)\|\vec{v}\|^2 - \|\vec{u}\|^2 - i \langle \vec{u}, \vec{v} \rangle -$$

$$+ i \langle \vec{v}, \vec{u} \rangle + i^2 \|\vec{v}\|^2) = \frac{1}{2}[-i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle})] =$$

$$z - \bar{z} = 2i \operatorname{Im}(z)$$

$$\stackrel{\downarrow}{=} \frac{1}{2}[-i \cdot 2i \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)] = \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \text{LS}$$

$$\text{PS} = \text{LS}$$